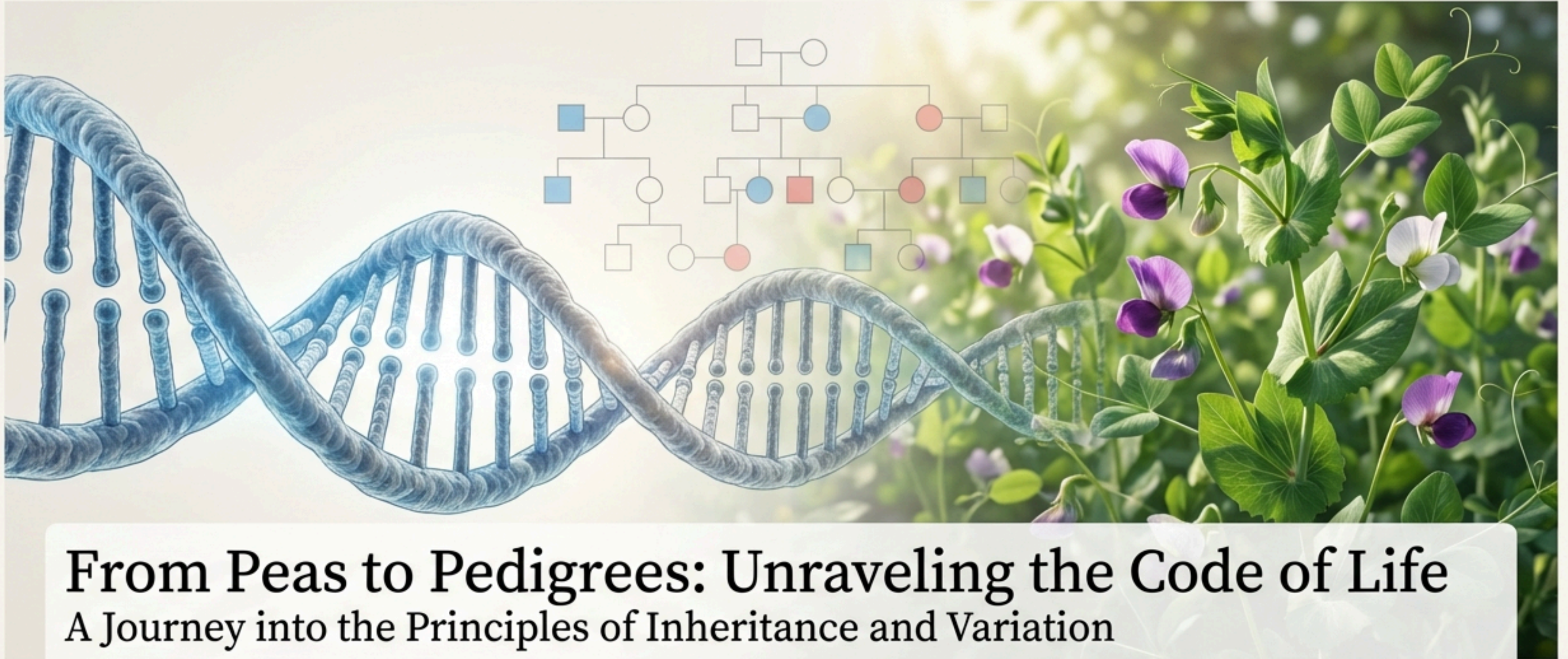




From Peas to Pedigrees: Unraveling the Code of Life

A Journey into the Principles of Inheritance and Variation

Why do offspring resemble their parents? (e.g., an elephant gives birth to a baby elephant)

Why are they not identical? (e.g., why siblings can look similar or very different)

Genetics: The science dealing with inheritance and variation.

Inheritance: The process by which characters pass from parent to progeny. It is the basis of **heredity**.

Variation: The degree by which progeny differ from their parents.

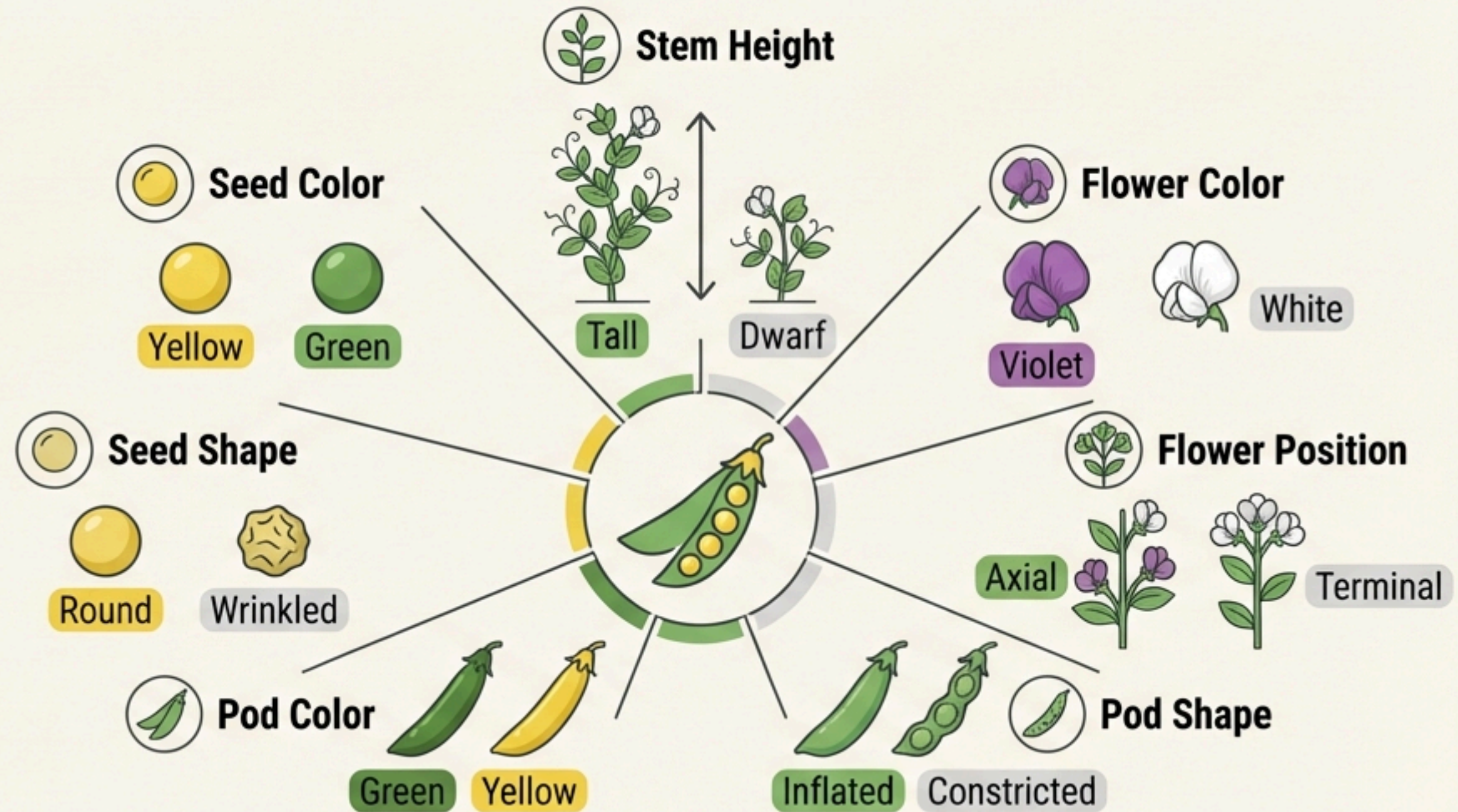


Gregor Mendel conducted hybridization experiments on garden peas for seven years (1856-1863), proposing the laws of inheritance in living organisms.

His Revolutionary Approach:

- **Statistical & Mathematical Logic:** For the first time, these were applied to problems in biology.
- **Large Sampling Size:** Gave immense credibility to his data.
- **Rigorous Verification:** Confirmed inferences over successive generations of his test plants.

The Father of Genetics: Mendel's Methodical Brilliance



Why the Garden Pea (*Pisum sativum*)?

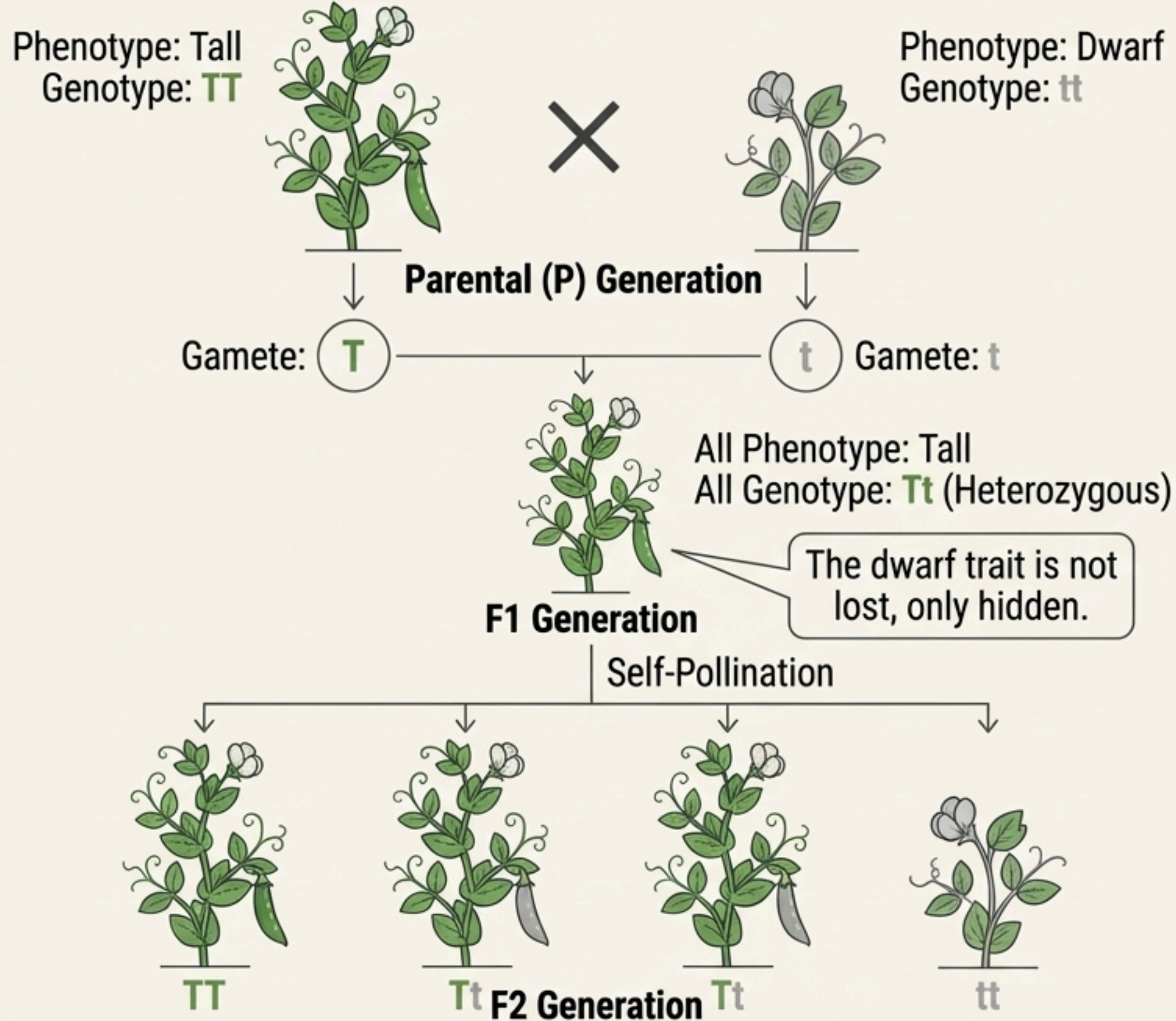

Easy to cultivate.


Short life cycle.


Produces many offspring.


Has clearly observable contrasting traits.

The Monohybrid Cross: Tracing a Single Trait



- **Phenotype:** The observable trait (Tall, Dwarf).
- **Genotype:** The genetic makeup (TT, Tt, tt).
- **Alleles:** Alternative forms of a gene (T and t).
- **Homozygous:** Identical alleles (TT, tt).
- **Heterozygous:** Dissimilar alleles (Tt).
- **F2 Phenotypic Ratio:**
3 Tall : 1 Dwarf
- **F2 Genotypic Ratio:**
1 TT : 2 Tt : 1 tt
- **Key Insight:** Traits do not blend. The recessive trait reappears unchanged in the F2 generation.

From Observations to Laws: Dominance & Segregation

The Law of Dominance

- (i) Characters are controlled by discrete units called **factors** (genes).
- (ii) Factors occur in pairs.
- (iii) In a dissimilar pair (heterozygote), one factor **dominates** the other (**recessive**).

Explains: Why F1 shows only one parental trait and the 3:1 ratio in F2.

The Law of Segregation

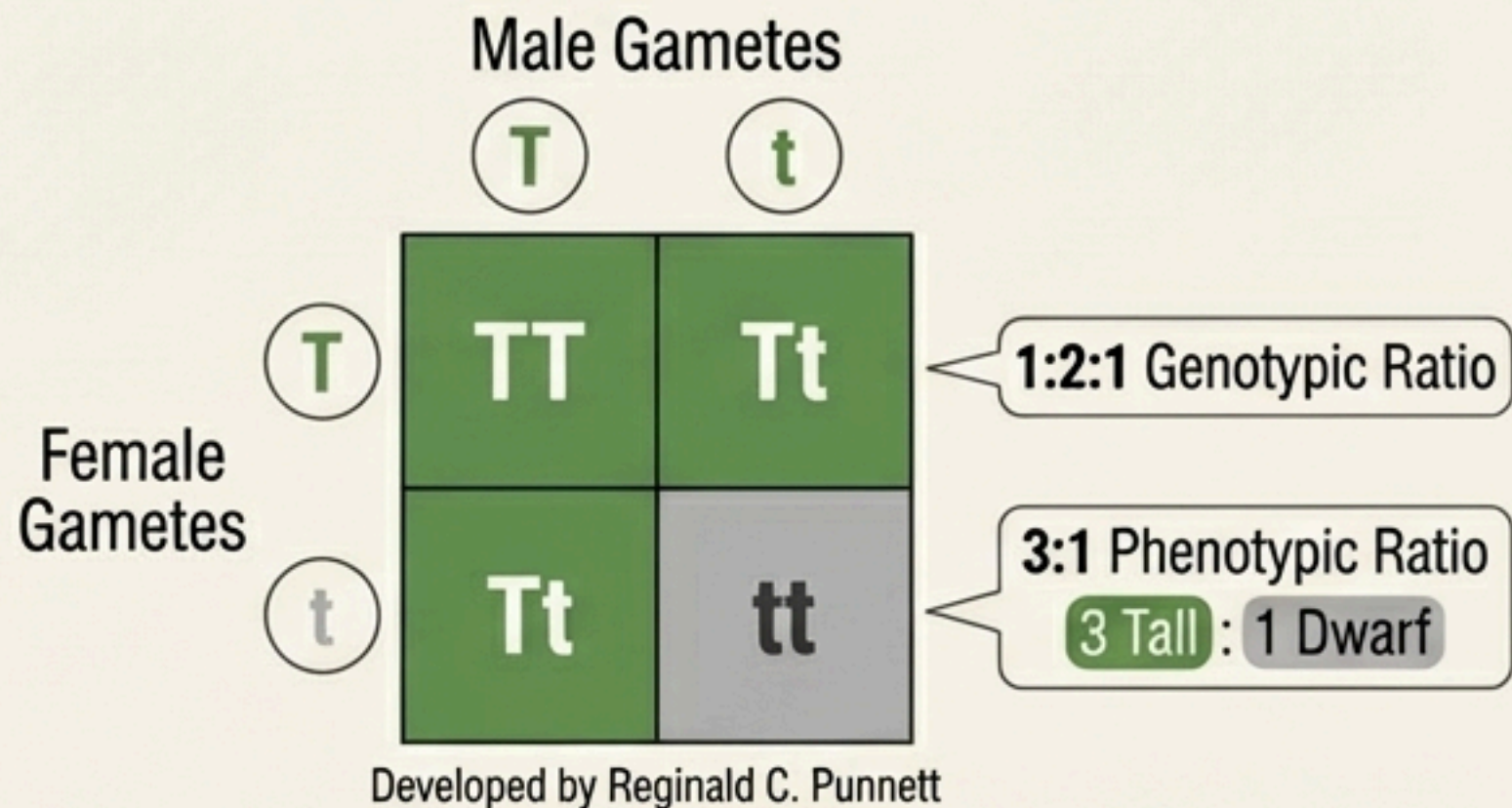
Alleles do not blend.

During gamete formation, the two alleles of a pair **segregate** from each other.

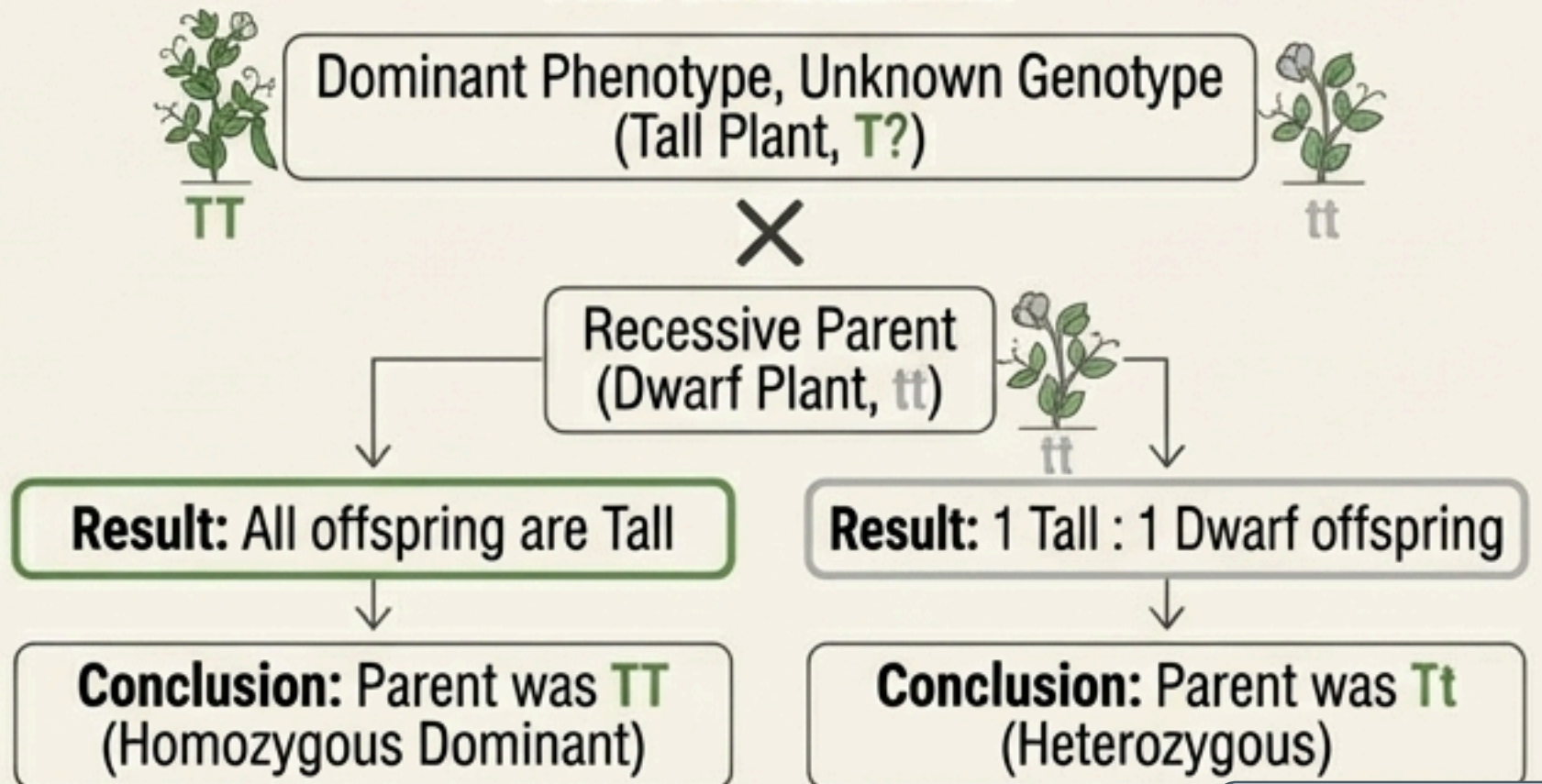
Each gamete receives only one of the two factors.

Key detail: A heterozygous individual (Tt) produces two types of gametes (T and t) in equal proportion.

The Punnett Square



The Test Cross

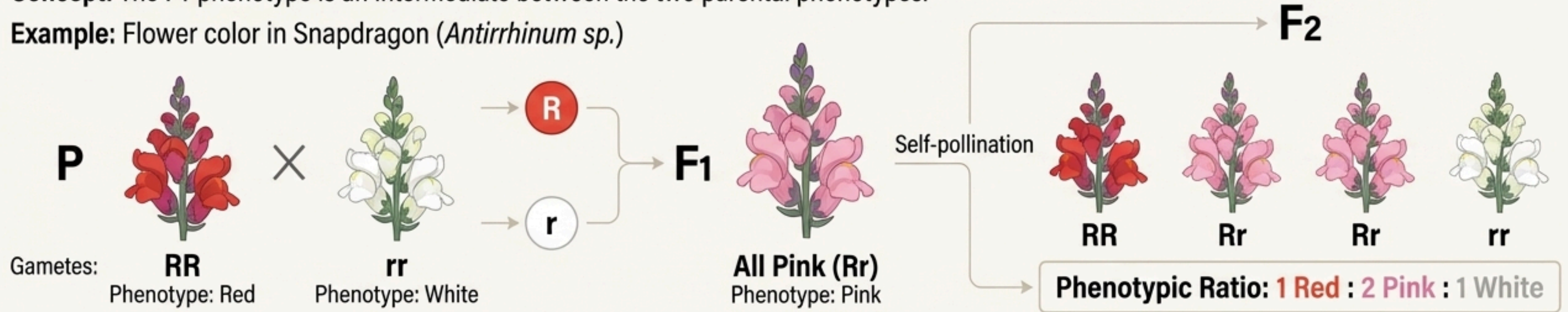


Expanding the Rules: Beyond Simple Dominance

Incomplete Dominance

Concept: The F1 phenotype is an intermediate between the two parental phenotypes.

Example: Flower color in Snapdragon (*Antirrhinum sp.*)



Co-dominance & Multiple Alleles

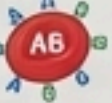
Both alleles in the heterozygote express themselves fully and simultaneously.

Example: Human ABO blood grouping, controlled by the gene I.

Three alleles exist: I^a , I^b , and i . This is an example of **Multiple Alleles**.

I^a and I^b are co-dominant, and both are dominant over i .

Human ABO Blood Group Genotypes & Phenotypes

Genotype	Resulting Phenotype (Blood Type)
$I^a I^a$ or $I^a i$	A 
$I^b I^b$ or $I^b i$	B 
$I^a I^b$	AB 
ii	O 

The Dihybrid Cross: Tracking Two Traits at Once

The Experiment

P Generation



Gametes: RY from one parent, ry from the other.



F₁ Generation:
All offspring are Round, Yellow (RrYy).

F₁ Generation

All offspring are Round, Yellow (RrYy).



Dominance is established:

Round > Wrinkled,
Yellow > Green.

F₁ Self-Cross: RrYy x RrYy

Inter:

RY

Ry

rY

ry

RY

RRYY



RRYy



RrYY



RrYy



Ry

RRYy



RRyy



RrYy



Rryy



rY

RrYY



RrYy



rrYY



rrYy



ry

RrYy



Rryy



rrYy



rryy



Phenotypic Ratio: 9 : 3 : 3 : 1

9 Round, Yellow 3 Round, Green 3 Wrinkled, Yellow 1 Wrinkled, Green

Key Takeaway: The segregation of seed shape is independent of the segregation of seed color. This is the **Law of Independent Assortment**.

The Physical Basis: Chromosomal Theory of Inheritance

Mendel's work, published in 1865, remained unrecognized for 35 years.

Rediscovery (1900): Advances in microscopy allowed scientists to observe chromosomes during cell division.

The Insight (1902): Walter Sutton and Theodore Boveri independently noted the striking parallels between the behavior of Mendel's "factors" and the behavior of chromosomes during meiosis.

GENES / ALLELES (Mendel's Factors)

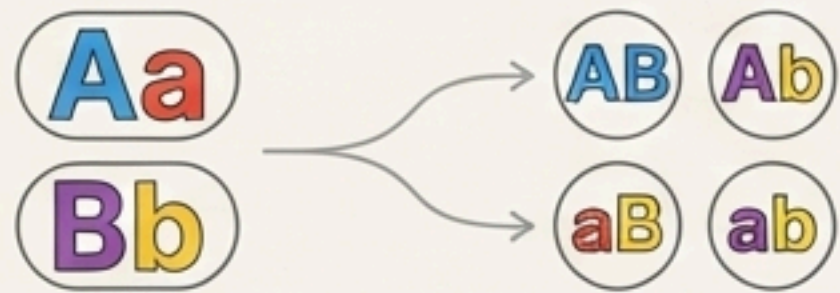
Occur in **pairs**.



Segregate during gamete formation, so each gamete receives only one.



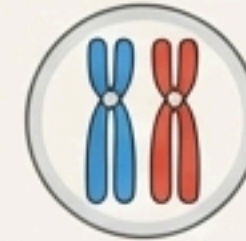
Independent pairs assort independently.



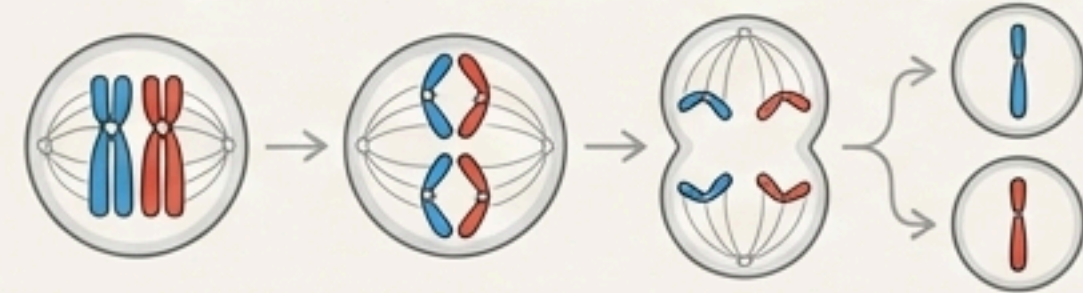
The Parallel Behavior

CHROMOSOMES (Observed in Meiosis)

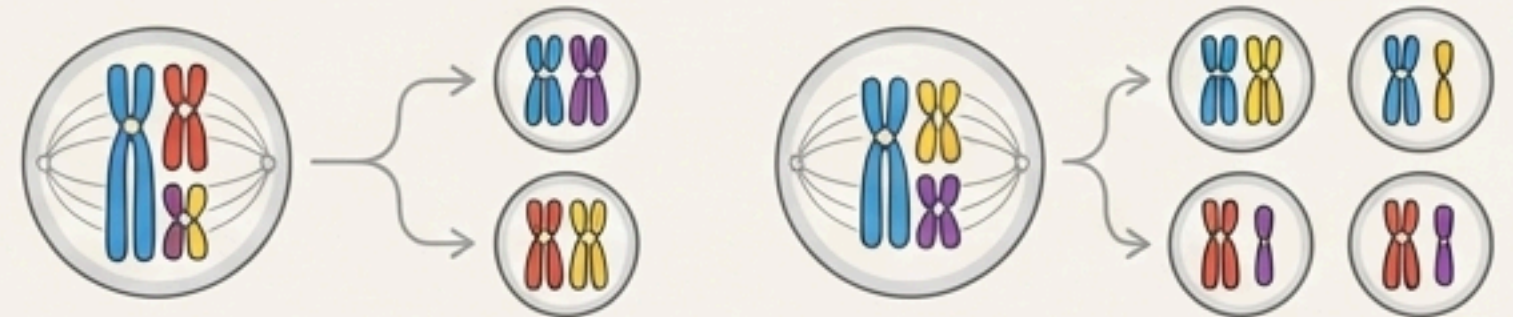
Occur in **homologous pairs**.



Homologous chromosomes **separate** during Meiosis I.



Independent pairs align and segregate independently at the metaphase plate.



The Chromosomal Theory of Inheritance

Sutton united chromosomal segregation with Mendelian principles, formally stating that **genes are located on chromosomes**.

Proof and a Puzzle: Morgan, Fruit Flies, and Linkage

The Scientist & The Model Organism



T.H. Morgan's work provided the experimental verification for the Chromosomal Theory of Inheritance.

Why the Fruit Fly (*Drosophila melanogaster*)?



Simple synthetic medium culture.



~Two-week life cycle.



Many observable variations.

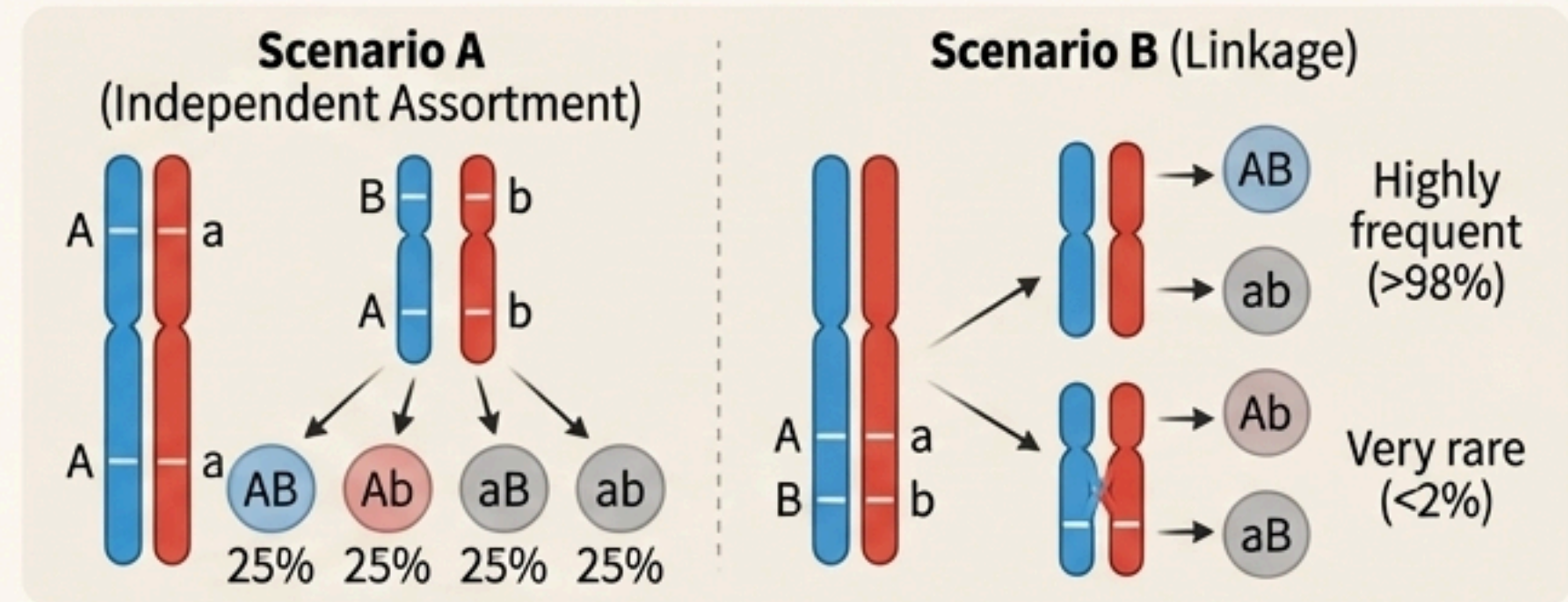


Single mating produces large progeny.

The Discovery of Linkage

Morgan's dihybrid crosses did **not** yield the expected 9:3:3:1 ratio.

Observation: The parental gene combinations (e.g., yellow body, white eyes) appeared far more frequently than non-parental combinations.



Conclusion: The two genes were physically linked on the same chromosome (in this case, the X chromosome).

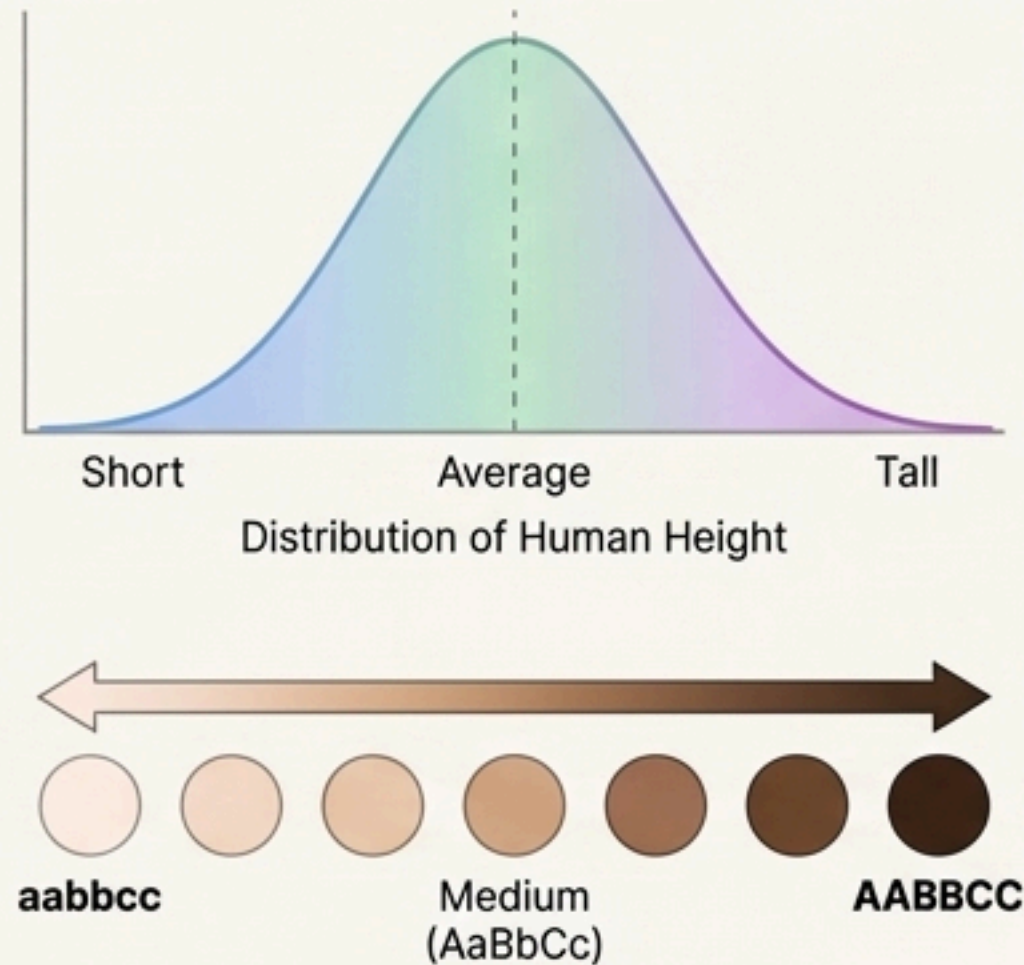
Key Definitions

- **Linkage:** The physical association of genes on the same chromosome, causing them to be inherited together.
- **Recombination:** The generation of non-parental gene combinations, caused by crossing over during meiosis.
- **Sturtevant's Insight:** Morgan's student, Alfred Sturtevant, realized that recombination frequency could be used to measure the distance between genes, creating the first **genetic maps**.

The Spectrum of Inheritance: From Skin Color to Sex

Polygenic Inheritance

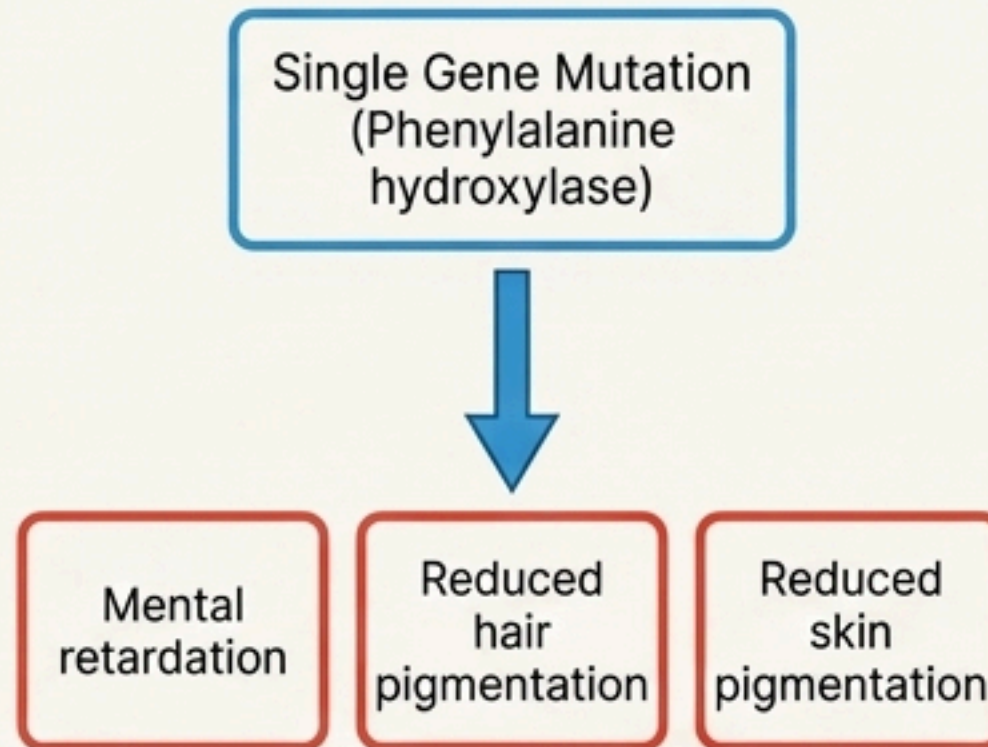
Traits controlled by three or more genes, creating a continuous gradient of phenotypes. Often influenced by the environment.



Pleiotropy

One gene influences multiple, seemingly unrelated phenotypic traits.

Phenylketonuria (PKU)



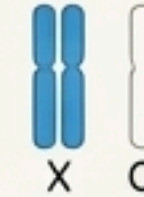
Sex Determination Mechanisms

XO Type

(Grasshopper)

Male: **XO**

Female: **XX**



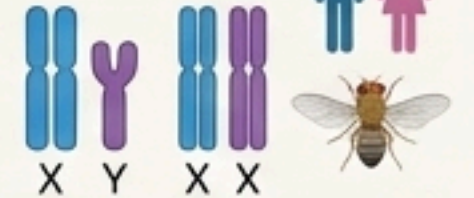
XY Type

(Humans, Drosophila)

Male: **XY**

(Male heterogamety)

Female: **XX**



ZW Type

(Birds)

Male: **ZZ**

Female: **ZW**
(Female heterogamety)



Haplodiploidy

(Honey Bee)

Female: **Diploid (32)**
from fertilized egg

Male: **Haploid (16)**
from unfertilized egg

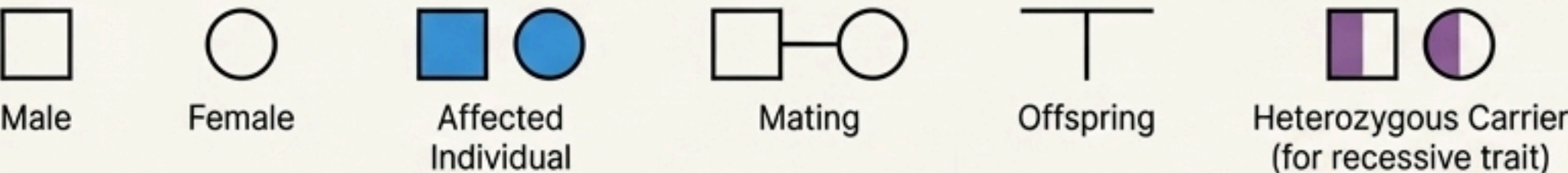


Tracing Our Traits: Pedigrees and Mutations

Pedigree Analysis: A Human Family Tree

Since controlled crosses are impossible in humans, pedigree analysis tracks the inheritance of a trait or disease through generations.

Symbol Key



Example Pedigrees

Chart A: Autosomal Dominant
(e.g., Myotonic Dystrophy)

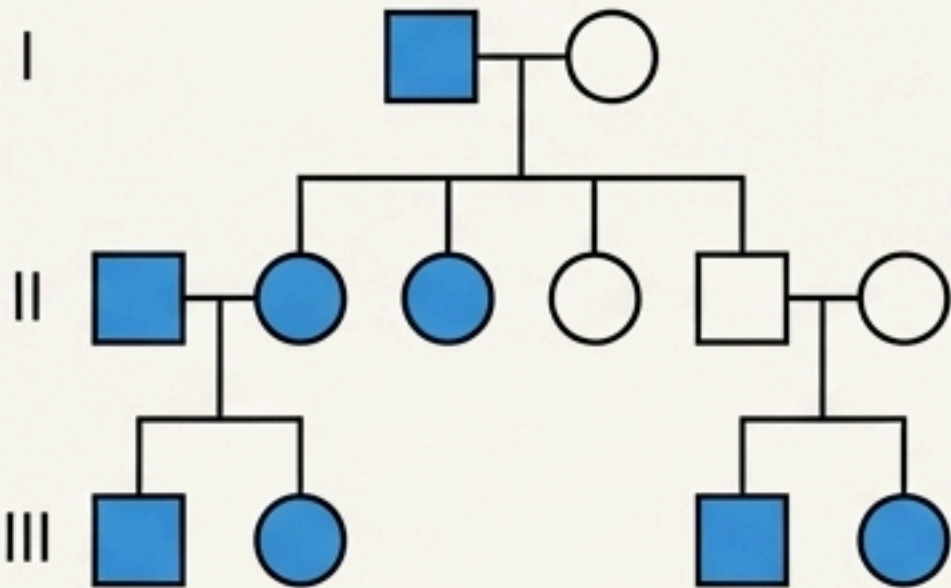
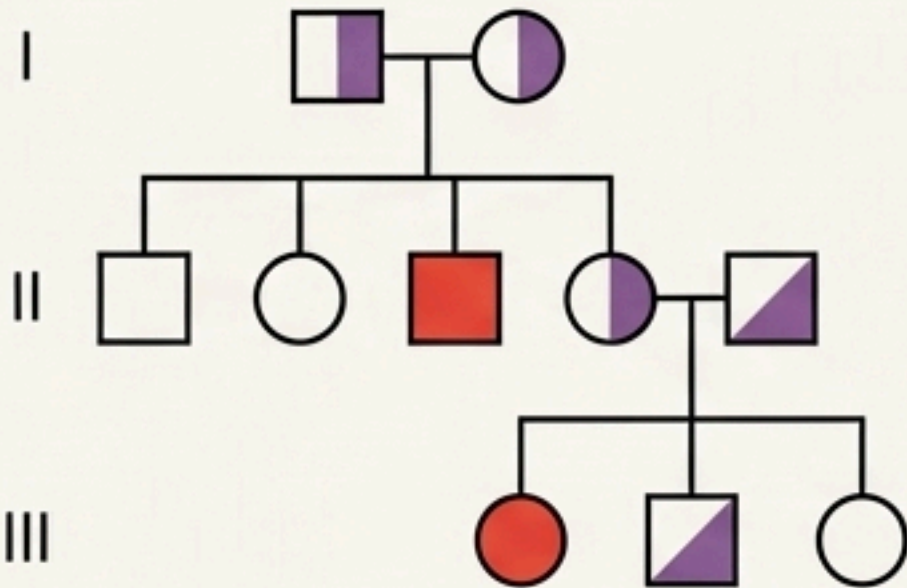


Chart B: Autosomal Recessive
(e.g., Sickle-cell Anaemia)

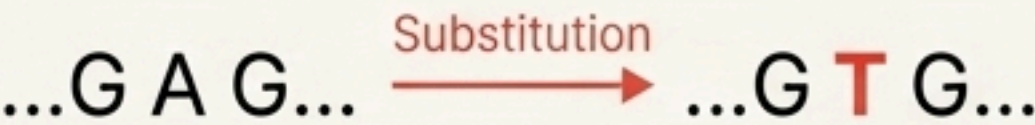


Mutation: The Source of Variation

An alteration of DNA sequences that results in changes in genotype and phenotype.

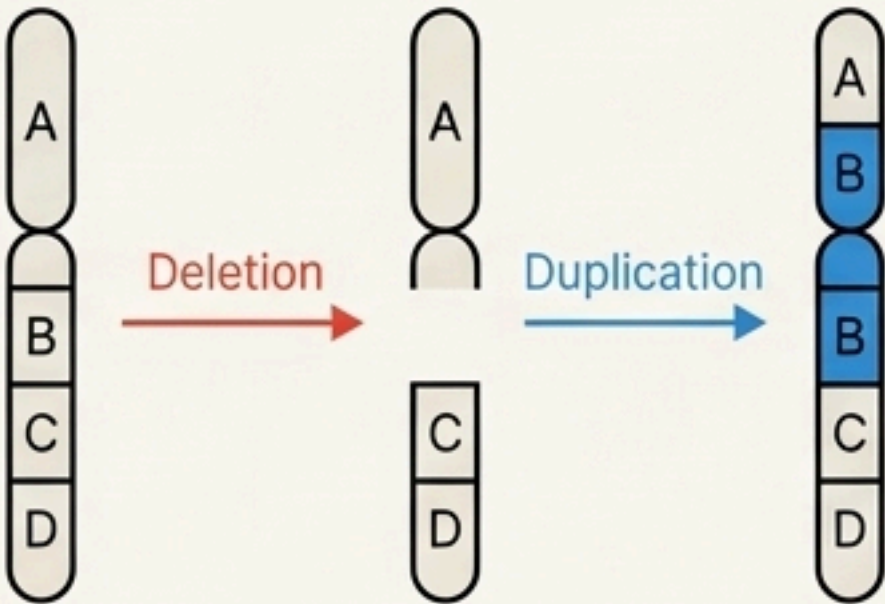
Mutation Types

Point Mutation

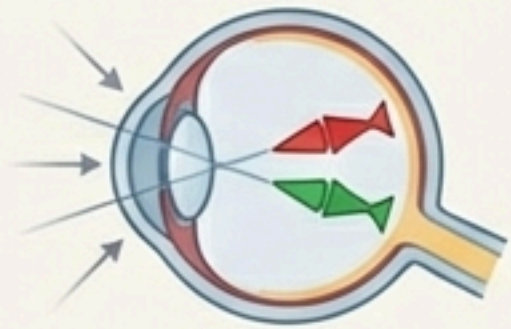


Example: Sickle-cell anaemia

Chromosomal Aberration



Mendelian Disorders: The Impact of a Single Gene



Colour Blindness

Type: X-linked recessive disorder.

Mechanism: Defect in red or green cone cells of the eye due to a **mutation on the X chromosome**.

Prevalence: ~8% of males, ~0.4% of females.

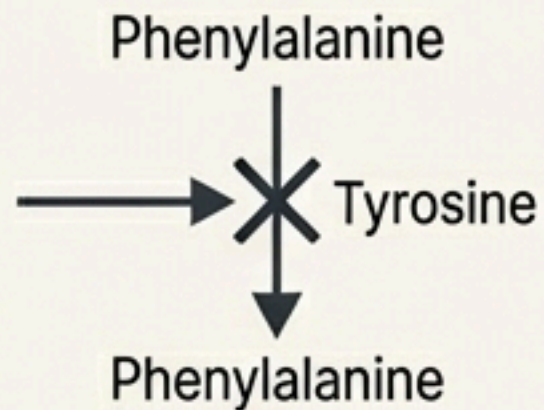


Haemophilia

Type: X-linked recessive disorder.

Mechanism: A single protein in the blood clotting cascade is defective, leading to non-stop bleeding.

Inheritance: Transmitted from a carrier female to male progeny.



Phenylketonuria (PKU)

Type: Autosomal recessive disorder.

Mechanism: Lack of the enzyme that converts phenylalanine to tyrosine. Phenylalanine accumulates, causing **mental retardation**.



Thalassemia

Type: Autosomal recessive disorder.

Mechanism: A **quantitative** problem: reduced synthesis of α or β globin chains, leading to **anaemia**.

Types: α -thalassemia (**chromosome 16**) and β -thalassemia (**chromosome 11**).

Case Study: Sickle-Cell Anaemia

Inheritance Pattern

Inheritance: Autosomal recessive trait controlled by alleles Hb^a and Hb^s .

$Hb^a Hb^a$: Normal $Hb^a Hb^s$: Carrier (Sickle-cell trait), generally unaffected. $Hb^s Hb^s$: Affected with sickle-cell anaemia.

The Molecular Defect (A Point Mutation)

Step 1: The Gene (DNA)

Normal β -globin Gene:
6th codon is **GAG**.



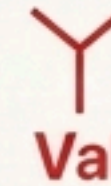
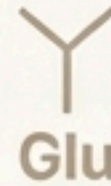
Substitution



Mutant β -globin Gene:
6th codon becomes **GUG**.

Step 2: The Protein (Amino Acid)

Normal Protein:
6th amino acid is
Glutamic acid (Glu).



Mutant Protein:
6th amino acid becomes
Val Valine (Val).

The Cellular Consequence

Under low oxygen, the mutant haemoglobin polymerizes, changing the Red Blood Cell's shape.

Normal RBCs

Biconcave disc shape



Sickled RBCs

Elongated, crescent shape



Chromosomal Disorders: When the Count is Wrong

Aneuploidy: Gain or Loss of Chromosomes

Cause: Failure of segregation of chromatids during cell division.

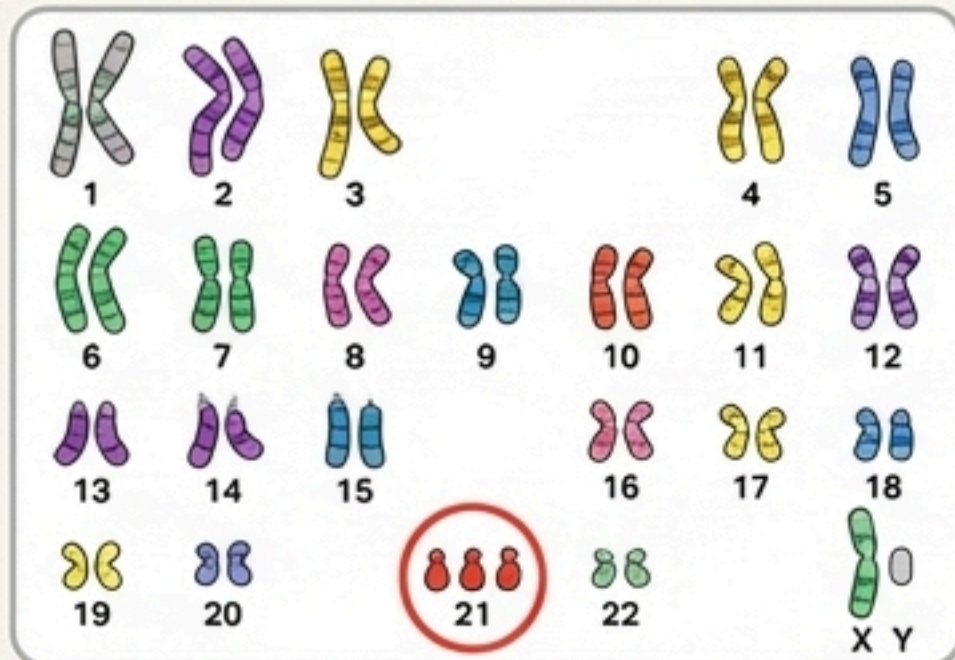
Types: **Trisomy:** Gain of an extra chromosome ($2n+1$).

Monosomy: Loss of a chromosome ($2n-1$).

Down's Syndrome

Cause: **Trisomy of Chromosome 21**
(Total chromosomes = 47).

Symptoms: Short stature, small round head, furrowed tongue, broad palm with a single palm crease, retarded physical and mental development.

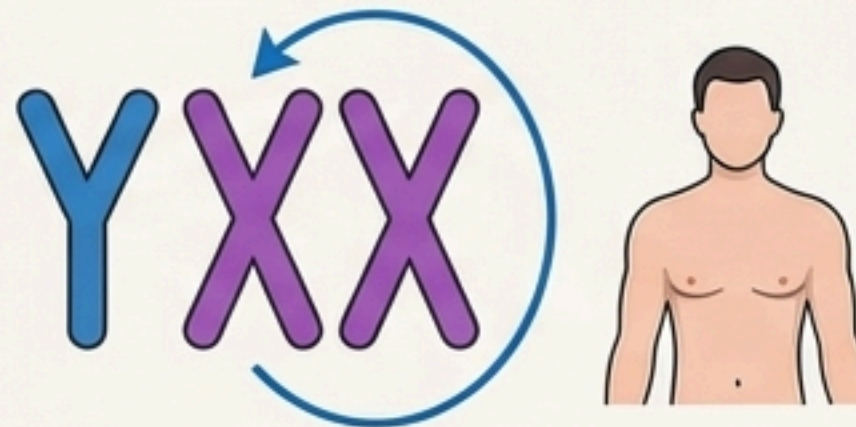


Klinefelter's Syndrome

Cause: Presence of an additional X chromosome in a male.

Karyotype: **47, XXY**

Symptoms: Overall masculine development, but with feminine characteristics like breast development (Gynaecomastia). Individuals are sterile.

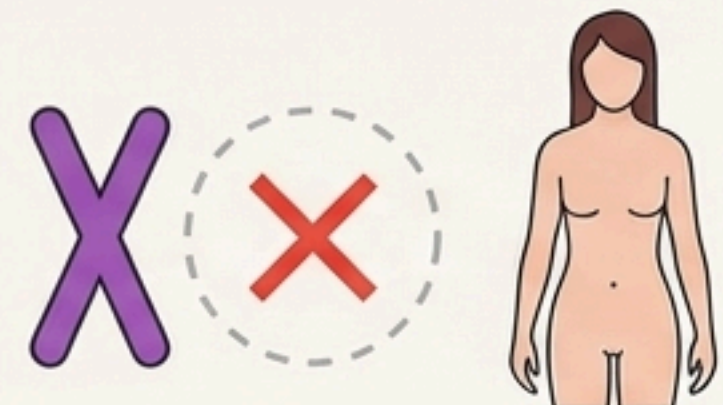


Turner's Syndrome

Cause: Absence of one X chromosome in a female.

Karyotype: **45, XO**

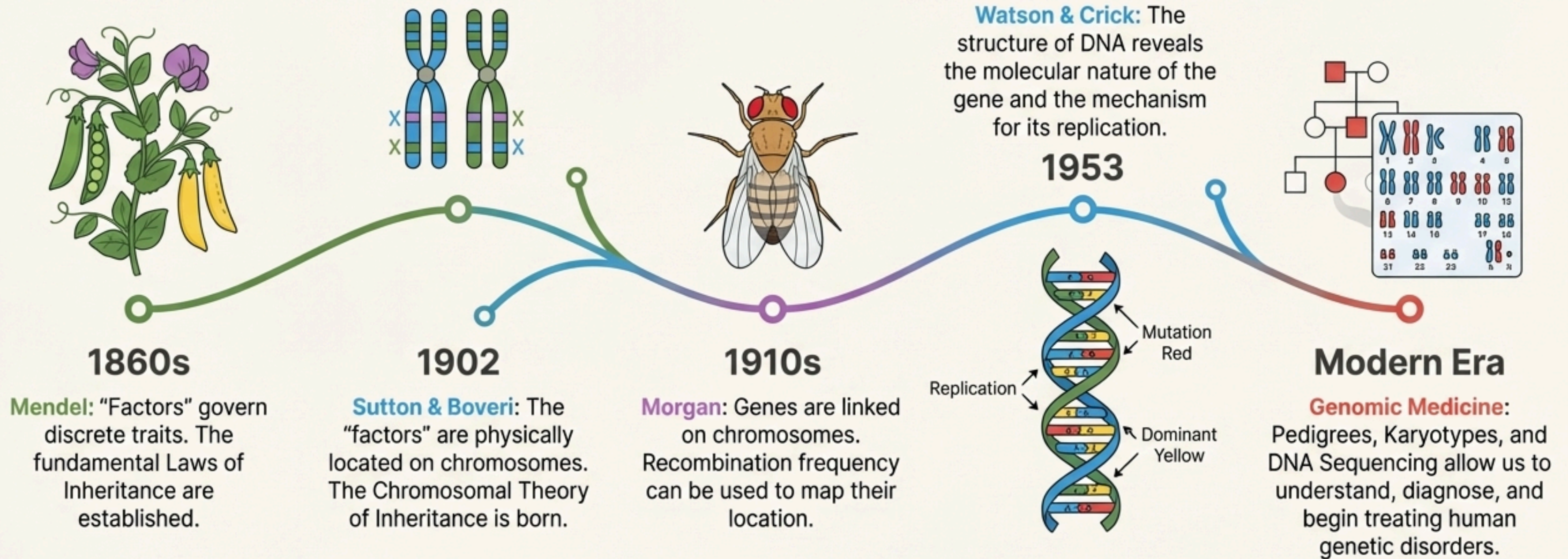
Symptoms: Females are sterile as ovaries are rudimentary; lack of secondary sexual characters.



A Framework for Genetic Disorders

Mendelian Disorders		Chromosomal Disorders	
Cause:	Alteration or mutation in a single gene .	Cause:	Absence, excess, or abnormal arrangement of one or more chromosomes .
Mechanism:	Follows principles of Mendelian inheritance (dominant, recessive, sex-linked).	Mechanism:	Arises from errors in cell division (e.g., nondisjunction leading to aneuploidy).
Analysis Tool:	Traced using Pedigree Analysis .	Analysis Tool:	Diagnosed by Karyotyping .
Nature of Problem:	A “qualitative” or “quantitative” protein issue.	Nature of Problem:	A large-scale error involving many genes at once.
Examples:	• Sickle-cell Anaemia  • Haemophilia  • Phenylketonuria  • Thalassemia 	Examples:	• Down's Syndrome  • Klinefelter's Syndrome  • Turner's Syndrome 

The Unifying Code: From Factors to Genomes



The **principles of inheritance**, first uncovered in a monastery garden, now form the foundation of **modern biology and medicine**. Understanding this code is not just an academic exercise—it is fundamental to understanding **human health, disease**, and life itself.